Trigonometric identities

1 $\dfrac{\sin x}{\cos x}$

2

a $\dfrac{1}{2}$ **b** $\dfrac{a}{c}$

3

a $\dfrac{8}{17}$ **b** $-\dfrac{12}{13}$

4

a $\dfrac{4}{3}$ **b** $\dfrac{2}{\sqrt{5}}$

5 $\csc\theta = \dfrac{1}{\sin\theta}$

6

a $\cot\theta$ **b** $\sin^2(\theta)$ **c** $\sec^2(x)$ **d** $\cot\theta$

e $\tan\theta$ **f** 1 **g** $\sin\theta$ **h** 1

i $\sin y$ **j** $\tan\theta - 1$ **k** $\sin^2(\theta)$ **l** $\csc\theta$

m $\sec\theta$

7

a 2 **b** $\dfrac{17}{31}$ **c** $\dfrac{11}{6}$ **d** $\pm\dfrac{1}{\sqrt{2}}$

8 $\dfrac{9}{2}$

9 $\dfrac{7}{4}$

10 $\dfrac{1}{11}$

11 $\dfrac{3}{8}$

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12

a $\frac{\sqrt{3}}{3}$ b $-\frac{\sqrt{6}}{4}$

13 Impossible. If $\tan\theta = 3.5$ then $\cot\theta = \frac{1}{3.5} \approx 0.29 \neq -3.5$.

Pythagorean identities

14

a Yes b Yes c No d No

15 Impossible

16

a $\sin\theta$ b $-\sin^2(\theta)$ c $1 - 2\cos\theta\sin\theta$ d $-\cos^3(\theta)$

e $10 - 6\cos x$ f 1 g $-\tan\theta$ h 1

i $\sin^2(\theta)$ j $-\cot\theta\cos\theta$ k $-\tan^2(x)$ l $\tan\theta - \cot\theta$

17

a $\tan^2(\theta)$ b $\cos^2(\theta)$ c $\cos^2(x)$ d $\csc^2(\theta)$

e $\sec^2(\theta)$ f $\tan^4(x)$ g $\tan^2(\theta)$

18

a $2\csc^2(\theta)$ b $\cot^2(x)$ c $\tan^2(x)$ d $\sec\theta\csc\theta$

e $\sec^2(x)$ f $\cot^2(x)$ g $2\csc\theta$ h 1

i 1 j $2\cos\theta\csc^2(\theta)$ k $-\cot x$ l $\cot\theta\cos\theta$

19 $a\sec\theta$

20 $\cot x = \pm\sqrt{\csc^2(x) - 1}$

21

a $\sin x = \frac{5}{13}$ b $\tan x = \frac{5}{12}$

22 $\cos\theta = -\frac{1}{2}$

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